

AN EGYPTIAN PATTERN IN THE LINEAR A LIBATION FORMULA

Summary

The votive inscriptions of Linear A share a formula that recurs at six Cretan sanctuaries over more than two centuries and remains unread. We propose three things about it, fixed by its distribution and not by a borrowed language: that its architecture is that of the Egyptian offering formula, and of no other contemporary tradition, measured position by position against 1,357 real instances of that formula; that U-NA-KA-NA-SI is the term for the offering; and that what can be predicted of a new vessel has a ceiling of 14% that belongs to the corpus. Over the eighteen non-accounting inscriptions preserving at least one anchor it establishes that the formula opens with a fixed word whose root is invariant and whose ending varies by sanctuary; that the fillers of the variable slots are hapax in 32 of 32 cases (probability 10^{-4}), so that the slots hold the open class of names and the anchors the closed class; that the one inscription with logograms, SY Za 2, fixes U-NA-KA-NA-SI as the term for the offering without etymology, since it is followed by the oil logogram. Compared as a structure with the contemporary votive formulae that are read, the Minoan template places its central invariant element between the names and the offering, as the Egyptian does. Two structural readings survive for the central anchor and two for the next; the datum that would separate them is absent from the corpus, and where it might appear is stated. A held-out test with baselines closes the paper: reconstructing each inscription's slots from the others recovers 8 of 57 elements (14.0%); an n-gram baseline does better at Top-1 (17.5%); both are bounded by the same fact, that the true filler is among the candidates in only 14.0% of slots. All converge near 23%, a ceiling of the corpus and not of any model.

1. A TEXT THAT REPEATS, AND THAT NOBODY READS

Around 1600 BC, at six mountain and cave sanctuaries of Crete, one and the same statement was cut on stone vessels, libation tables and ladles. It is recognisable at a glance: it opens with A-TA-I-*301-WA-JA, goes on with a word that changes from vessel to vessel, and continues with JA-SA-SA-RA-ME, U-NA-KA-NA-SI, I-PI-NA-MA and SI-RU-TE, almost always in that order and almost always the same. It is the only text of Linear A that repeats, and therefore the only one with something like a visible grammar. A century after its identification (Evans; Duhoux 1992; Finkelberg 1990-91; Davis 2013, 2014; Karnava 2016; Petrakis and Steele 2025), nobody knows what it says.

What has been tried is to read it with a borrowed language. Finkelberg (1990-91) equated (J)A-SA-SA-RA-ME with Luwian *ishassaras*, “lady”, and hence a deity, the most cited proposal; Facchetti and Negri (2003) read the opening as a divine name; others have read the formula from Semitic or from Greek. None of those readings can be compared with the others, because each starts from a different language and none says what observation in the corpus would refute it. For a century the formula has been the place where every hypothesis about Minoan is put to the test, and where none has been able to lose.

This paper asks the question the other way round: not what the formula says in another language, but what it says by itself. A text that repeats with slots fixes, by its distribution alone, what each position does: which is fixed and which open, which carries names, which carries the offering, which varies with the place. That can be established without reading a word, against a null that says how much of it chance would produce, and it can be compared with the votive formulae of

the same period that are read. It can be taken further: a fixed template predicts, and one can measure how much of a new vessel it recovers from the others, and why not more.

Three things result. First, the Minoan formula has the architecture of the Egyptian votive formula, not that of the Semitic or the Greek: a central invariant element between the names and the offering, and a fixed formula before the dedicant. Second, U-NA-KA-NA-SI is the term for the offering, fixed by a logogram and not by an etymology: on the one vessel that carries logograms it is followed by the sign for oil. Third, the template has a prediction ceiling of 14%, and that ceiling belongs not to the method but to the corpus: 72% of the fillers occur on no other vessel, and no procedure, ours or the reference n-gram, recovers what is not there.

With that, of the dozen readings in circulation two remain per position for the two central anchors, and one can say which inscription would be needed to decide between them: a datum that is not in the corpus, and that an excavator would recognise if it appeared. That is what this paper offers the field: not one more reading, but the exact shape of the one that is missing.

2. DATA AND METHOD

Eighteen non-accounting inscriptions preserve at least one anchor: IO Za 2, 6, 9, 14, 15, 16 (Iouktas); KN Za 10 (Knossos); KO Za 1 (Kophinas); PK Za 11, 12, 27 (Palaikastro); PL Zf 1 (Platanos); PS Za 2 (Psychro); SY Za 2, 3 (Syme); TL Za 1 (Tylissos); VRZ Za 1 (Vrysinas); AP Za 2 (Apodoulou). Corpus: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer, with sign verification in SigLA.

A caveat on the size of the votive corpus. Classifying by support inflates it: of the 107 stone vessels inscribed in Linear A, 33 are accounting documents and only 15 carry the formula, and the 33 cluster at Zakros, with up to twelve quantities on a single piece. The votive corpus of Linear A is seventeen documents, not a hundred and seven, and that sets the statistical power of everything that follows (Acedo 2026b).

The measurements are distributional: the position of each anchor, the hapax rate of the fillers, co-occurrence with logograms, and the variation of the endings cross-tabulated with site. Where a null is needed, it is a permutation of labels within strata.

Linear A sign-groups are read here by applying Linear B values to Linear A signs, the approach recent scholarship regards as methodologically legitimate for the study of Linear A (Steele and Meißner 2017), and are given in capitals with hyphens, the convention of Younger (2024) and of Salgarella and Petrakis (2025). The eighteen inscriptions, aligned by slot, are given in Fig. 1.

[Fig. 1 near here]

3. THE TEMPLATE, FIXED BY DISTRIBUTION

3.1 The opening is a root with a paradigm

A note on notation first. The transliteration used splits *301 off as a separate unit, which conceals the opening anchor; here it is restored to its word. The sign *301 is left without a phonetic value, following GORILA and the edition consulted. Part of the decipherment literature assigns it the value JO and therefore writes A-TA-I-JO-WA-JA where this paper reads A-TA-I-*301-WA-JA. The difference is not one of reading the tablets but of assigning a value to a sign,

and it affects equally all sixteen inscriptions in which the opening occurs. Nothing that follows depends on that assignment, since *301 is constant across all variants and what is measured is the variation of the endings. So restored, the opening (A) occurs in sixteen inscriptions and always in first position, with an invariant root (A-)TA-I-*301 and six endings: -WA-JA (eleven times), -WA-E, -U-JA, -U-TI-NU, -TI and -DE-KA.

The endings group by sanctuary: Syme five of five with -WA-JA; Iouktas three with -WA-JA and one with TA-NA-I-*301-U-TI-NU; Palaikastro one -WA-E and one -WA-JA; Apodoulou and Psychro with forms of their own. Nor do they covary with the ending of the toponym that follows (-WA-JA occurs with -tu, -TE, -a and -mi), so the variation is one of local convention and not of grammatical agreement (p from 0.011 to 0.16 once site is controlled).

This is the corpus's best candidate for an inflectional or derivational paradigm on a single root, and it is stated as such without proposing what distinguishes each ending. The internal structure of this opening was analysed by Finkelberg (1990-91), and priority belongs to her: she observes that while the beginning and the end of the formula vary, its middle, the sequence -i-*301-, is stable, and concludes that it cannot be a single word but two semantic units, the first ending in -i, which she proposes to read as an element of nominal or verbal inflection. She further shows that the three endings are neither forms of one word (the element u/wa is absent from -TI, the element ti from -WA-JA) nor three distinct words, and describes them as conglomerates of elements. What is added here is the measurement of what the variation depends on: the endings group by sanctuary and do not covary with the ending of the toponym that follows, so that the variation is one of local convention and not of grammatical agreement.

3.2 The slots hold the open class

The slot preceding the central anchor has ten fillers at six sites, and holds toponyms identifiable through Linear B (JA-DI-KI-TU and A-DI-KI-TE = Dikte; I-DA-A and I-DA-MI = Ida) sometimes followed by two further words. The following slots have fourteen fillers.

All thirty-two fillers are hapax: they occur once in the whole corpus. Under the corpus hapax rate (0.76), the probability of thirty-two hapax in a row is of the order of 10^{-4} . The conclusion is structural and depends on no reading: the variable slots hold names and places, and the anchors the closed class.

3.3 SY Za 2 fixes the term for the offering (Fig. 2)

[Fig. 2 near here] A single inscription of the series carries logograms, and it decides one position. SY Za 2 writes “JA-SU-MA-TU OLIV | U-NA-KA-NA-SI OLE”: anchor D is followed by the logogram for oil, and the word preceding it carries that for olives.

That fixes D as the term for the offering, with no recourse to etymology. It is, so far as I can tell, the only point of the formula whose function is established by internal evidence rather than by comparison with a candidate language. The observation about this inscription is due to Petrakis and Steele (2025); what is added here is its distributional consequence.

3.4 What JA-SA-SA-RA-ME cannot be

The central anchor occurs thirteen times at six sanctuaries, with variants of prefix (ja-, a-, none) and of ending (-ME, -MA, -MA-NA), and never carries a logogram or a quantity. That absence,

across thirteen occurrences and with SY Za 2 showing that the formula does admit logograms, rules out two functions: JA-SA-SA-RA-ME designates neither the offering nor the object. And its invariance across sanctuaries rules out two more: it is not the dedicant's name (which would be hapax) nor the sanctuary's (which would vary by site). This agrees with the conclusion of Petrakis and Steele (2025), who find that the recent evidence does not support the reading of (J)A-SA-SA-RA-ME as a theonym.

3.5 The resulting template

[A: fixed root + local ending] [place, and up to two names: hapax] [C: JA-SA-SA-RA-ME, no logogram] [D: U-NA-KA-NA-SI, admits a logogram of offering] [E: I-PI-NA-MA] [F: SI-RU-TE] [name or second form of the opening: hapax, optional]

With a variant Younger called “abbreviated”, at Syme: opening, word with OLIV, D with OLE, and logograms in place of anchors E and F.

3.6 Separable morphology

Three elements separate by distribution and not by etymology.

I-NA-: two root/prefixed pairs in the whole corpus, both in formula context and both on a place name: I-NA-I-DA against I-DA (Ida), and I-NA-JA-PA-QA against JA-PA-QA, these two at the same sanctuary (PK Za 11 and 12). A candidate for a marker of place: located, not read. Younger discusses the prefix I- as a case marker; I-NA- is a longer form with the same distribution.

ja- / a-: eleven pairs with a- in the whole corpus (A-RI-JA / RI-JA, A-KI-RO / KI-RO, A-MI-TA / MI-TA), and the alternation JA-SA-SA-RA-ME / A-SA-SA-RA-ME / SA-SA-RA-ME. Its frequency is 7 per thousand at Haghia Triada against 140-240 per thousand at the sanctuaries: it is a feature of genre, not of grammar.

U-TI-NU: occurs only on IO Za 11, and as the ending of TA-NA-I-*301-U-TI-NU (IO Za 6) and of TA-NA-RA-TE-U-TI-NU (IO Za 2). The two Iouktas compounds share the frame TA-NA-[X]-U-TI-NU. It is a separable element that the script joins to the preceding word without a divider, and it has a consequence for the template: the tail filler of IO Za 2 is not a name but a second form of the opening, so that at Iouktas the formula closes as it opens.

4. THE EGYPTIAN PATTERN

The template of §3 can be compared, as a structure and without reading a word of it, with the votive formulae of the same period that are read. Four traditions are relevant by contact and date: the Egyptian offering formula, the Anatolian dedications in Hittite and Hieroglyphic Luwian, the Semitic dedications of Ugarit and Phoenicia, and, later, the Greek dedication. Each has a fixed order of its constituents (Table 1).

Table 1. Constituent order of four votive traditions against the Minoan template.

tradition	order	shares with Minoan
Egyptian, <i>htp-dj-nsw</i> (Barta 1968; Gardiner 1957, 170-173)	[fixed opening formula] [deity] [offerings] [beneficiary]	fixed opening before the dedicant; invariant element before the offering
Anatolian, Hittite votive texts and	[dedicant and title] [object] [verb]	dedicant before the invariant

tradition	order	shares with Minoan
Luwian dedications (de Roos 2007; Hawkins 2000)	[deity in the dative], or [deity] [dedicant] [verb]	element and the offering
Semitic, Ugaritic and Phoenician dedications (Pardee 2002)	[dedicant] [verb] [object] [to the deity]	none: the deity closes the text
Greek, archaic dedications (Lazzarini 1976)	[dedicant] [verb] [deity in the dative]	none: the deity closes the text

The Minoan template has two structural traits. The first is the order *names* → *invariant element* (*C*) → *offering* (*D*): the dedicant and place come first, the fixed central element next, the offering after it. That order is shared with the Egyptian and the Anatolian formulae and excludes the Semitic and the Greek, which close with the deity. The second is the fixed opening before the dedicant: A-TA-I-*301-WA-JA stands before any name in every complete inscription (Fig. 1). Among the four traditions only the Egyptian opens so: *ḥtp-dj-nsw*, “an offering which the king gives”, precedes the deity and the beneficiary in every instance of the formula from the Old Kingdom onward (Barta 1968). Hittite votive texts open with the dedicant or with the deity (de Roos 2007); Ugaritic and Greek dedications open with the dedicant.

The only tradition that matches both traits is the Egyptian; the Anatolian matches one. That is what the title claims, and it is stated with its measure: two traits of two against one of two, on eighteen fragments, with the fixed opening as the decisive trait. The claim is about the architecture of the formula, not about its language, and it is compatible with the phonotactic result that Minoan resembles none of thirteen candidate languages (Acedo 2026a).

The comparison can be made position by position, and not only as a description. The offering formula is extracted from the AES corpus (Schweitzer 2021, from the Thesaurus Linguae Aegyptiae): 1,357 instances of *ḥtp-dj-nswt* with their constituents tagged by that corpus (divine names, epithets and titles, offering substantives). For each instance the relative position of each constituent is computed (0 at the start of the formula, 1 at its end), and the same is done for the eighteen Minoan inscriptions.

Table 2. Relative position of each constituent, Minoan and Egyptian.

Minoan element	relative position	Egyptian constituent	relative position
A, A-TA-I-*301-WA-JA	0.04	the formula itself	0.00
C, JA-SA-SA-RA-ME	0.41	offering	0.41
fillers (names)	0.52	—	—
D, U-NA-KA-NA-SI	0.75	deity	0.62
F, SI-RU-TE	0.83	title, epithet	0.69

The skeleton is the same: a fixed opening at the very start in both, invariant elements in the middle third, closure in the last third. The content of the positions is not. In the Egyptian formula the offering precedes the deity (0.41 against 0.62); in the Minoan template the term for the offering is D, fixed by the logogram that follows it on SY Za 2, and it stands where the Egyptian puts the deity. The coincidence is therefore of architecture and not of constituent order, and the paper claims no more than that. One consequence is worth stating: a reading of C as a deity, which is what the Luwian equation proposes, gains no support from the Egyptian parallel, since

in the Egyptian formula that position belongs to the offering and in the Minoan the offering is fixed one position later.

Three cautions belong here. First, on precedent: the formula has been compared before, but always through a candidate language or a text read in one, with Linear B and Indo-European dedications (Facchetti, Negri 2003), with Hurrian and Anatolian material, and with the Orphic hymns on the basis of SY Za 2; Davis (2013) studied its word order to infer the type of language behind it. A comparison of its architecture, as a structure and without reading it, with the Egyptian offering formula I have not found in that literature nor in the bibliography of Karnava (2016). If a reviewer knows one, it belongs here. Second, on function: *htp-dj-nsw* is a funerary formula, and the Minoan text is a sanctuary dedication. What is compared is the architecture of offering texts, not their use; a shared architecture across different uses is, if anything, the more notable. Third, on interpretation: the most economical account of a shared architecture is contact, and the contact is documented for these very centuries by the Keftiu of the Theban tombs, the Kom el-Hetan list and the Minoan paintings at Avaris. That an administration in Crete adopted the shape of an Egyptian formula for its dedications is what such contact would produce, without adopting the language; the phonotactic result cited above says the language was not adopted. This is offered as the reading of the result, not as part of it.

4.1 The alignment, position by position, against 1,357 real offering formulae

The comparison above is between descriptions. It can be made between texts. From the AES corpus of the Thesaurus Linguae Aegyptiae (Schweitzer 2021), 1,357 instances of the *htp-dj-nswt* formula were extracted with the corpus's own part-of-speech labels (entity_name for deities, epitheton_title for titles, substantive for offerings), and the mean relative position of each element was computed on a 0-to-1 scale from the start of the formula to the end of the text. The same was done for the eighteen Minoan inscriptions with the slots of §3.

Minoan	position	Egyptian	position
A (A-TA-I- *301-WA-JA)	0.04	formula (<i>htp-dj-nswt</i>)	0.00
C (JA-SA-SA- RA-ME)	0.41	offering (substantives)	0.41
fillers (names)	0.52		
D (U-NA-KA- NA-SI)	0.75	deity	0.62
F (SI-RU-TE)	0.83	title, epithet	0.69

Two things follow, and the second matters more. The skeleton is the same, measured and not described: a fixed formula opens both traditions at position zero, the invariant elements sit in the middle third, and the closing elements in the last. But the content of the positions is not the same. In the Egyptian formula the offering precedes the deity (0.41 against 0.62). If positional correspondence were read as identification, the Minoan offering would be C and the Minoan deity would be D. The corpus says the opposite: D is the term for the offering, because SY Za 2 follows it with the oil logogram, and C never carries a logogram in thirteen occurrences.

The claim of this section is therefore about the skeleton and not about the content: the Minoan formula has the architecture of the Egyptian one and **places its offering where the Egyptian**

places its deity. That is a stronger statement than a shared outline, and it bears on the readings. A reading of C as a deity, which is what the Luwian equation proposes, cannot draw support from the Egyptian parallel, because in the Egyptian formula that position is the offering’s and in Minoan the offering is fixed one slot later.

The result has a consequence for the most cited reading. The equation of (j)A-SA-SA-RA-ME with Luwian *ishassaras*, “lady”, places a deity in slot C. Under the Anatolian order that is where the deity may stand; under the Egyptian order the deity stands after the fixed opening and before the offering, which is also C. The reading therefore survives the order test in both, but it loses the second trait: no Anatolian formula opens with a fixed element before the dedicant, and the Minoan does. The Luwian reading is not refuted by the architecture; it is deprived of the structural support it has been given, since the architecture points elsewhere.

5. WHAT THE TEMPLATE PREDICTS, AND WHAT NOTHING CAN PREDICT

Everything above describes the corpus. This section tests prediction on material that was set aside, following the evaluation practice of textual restoration (Sommerschild et al. 2023): a held-out test, several baselines including the field’s standard n-gram, and accuracy reported at several ranks rather than at the first alone.

Design. The eighteen inscriptions in which the opening anchor can be located are indexed by position relative to that anchor. For each slot, the whole inscription containing it is withheld and the element is predicted from the seventeen others by what occupies that relative position, weighting a candidate by how many other elements its source inscription shares with the one being reconstructed. The anchor position is excluded from scoring, since it is what the alignment uses; an early run that did not exclude it scored 20.2% and owed eleven of its seventeen hits to that position alone.

Results, with baselines and at several ranks:

[Fig. 3 near here]

Table 3. Held-out recovery by rank (Fig. 3).

	Top-1	Top-3	Top-5	Top-10	Top-20
this predictor	**14.0%**	14.0%	14.0%	14.0%	14.0%
n-gram baseline (predict from the preceding element)	**17.5%**	19.3%	21.1%	22.8%	22.8%
always guess the commonest element	8.8%	19.3%	21.1%	22.8%	24.6%

Two things must be said plainly, and the second explains the first.

The n-gram baseline beats this predictor at Top-1, 17.5% against 14.0%. The standard baseline of the field is better than the method proposed here, and reporting otherwise would require omitting it.

And this predictor's accuracy does not improve with rank at all, which is not how a predictor behaves unless it has run out of candidates. It has.

slots evaluated	57
the true element is available among the candidates for that position	8 (**14.0%**)
the true element occurs in no other inscription of the formula	41 (**71.9%**)

The ceiling of this method is 14.0%, and its accuracy is 14.0%. It recovers every slot the structure allows it to recover. In the remaining 86% the true element is not available to be proposed, and in nearly three quarters of cases because it is a hapax.

That is the result, and it confirms section 3.2 from the other side. The slots hold the open class: all thirty-two of the formula's fillers are hapax. One cannot predict the filler because the filler is new each time; one can predict the template because the template recurs. The eight hits are JA-SA-SA-RA-ME, I-PI-NA-MA and SI-RU-TE, which are exactly the three elements section 3 identifies as fixed by distribution. The same conclusion is reached twice by independent routes.

One figure is worth stating for anyone who attempts this corpus with a stronger method. All three approaches converge near 23% as rank grows, which is approximately the share of slots occupied by elements that occur more than once. The ceiling is a property of the corpus, not of the models: no method that works by recognising repetition will exceed it, however sophisticated, because 72% of the material to be recovered occurs exactly once.

What would refute this. New formula inscriptions raising the share of repeated fillers, which would lift the ceiling and make the comparison between methods meaningful; or the hits ceasing to concentrate on the fixed positions, which would sever the connection with section 3.

Calibration. The instrument was measured on accounting documents where no template exists: it reports a false positive in 5.0% of 120 trials at the 5% threshold and 0.83% at 1%. The permutation null for this test, shuffling elements within each inscription, gives 3.5%.

6. THE TWO READINGS THAT SURVIVE, AND THE DATUM THAT WOULD SEPARATE THEM

For each position, the functions a votive formula can assign to it are enumerated, drawing on the formulae of Egypt, Anatolia, Ugarit and Greece that are read, and those that contradict a measured datum are discarded.

6.1 The central anchor (C)

hypothesis	predicts	does it hold?
C1: name or title of the receiving deity	invariant across sites; no logogram; after the names and before the offering	yes throughout
C2: verb of dedicating	invariant; after the dedicant; might take an object after it	compatible , but then D would be the object and C-D would be verb plus object
C3: word for offering or object	should admit a logogram at some point	no: 0 of 13. Discarded

hypothesis	predicts	does it hold?
C4: dedicant's name	should be hapax	no: 13 near-identical occurrences. Discarded
C5: sanctuary's name	should vary by site	no: the same form at five sanctuaries. Discarded

The datum that would separate C1 from C2: if C ever appeared without D and with a logogram of offering after it, it would be a verb with an object; if it appeared in a non-votive text with a name before it and nothing after, it would be the deity. Neither occurs in the present corpus. The place where it might is the Knossos sceptre, a ritual text without numerals.

6.2 The offering anchor (D)

hypothesis	predicts	does it hold?
D1: term for the offering	admits a logogram; follows the deity	yes: OLE on SY Za 2; always after C
D2: verb of giving or offering	follows the recipient in the Egyptian and Anatolian formulae; may take an object	yes as well
D3: dedicant's title	should vary or be hapax	no: 9 near-identical at 5 sites. Discarded
D4: case marker or postposition	would not take a logogram of its own	no: OLE follows it. Discarded

D1 and D2 are not exclusive: a verb “to libate” and a noun “libation” share a root in many languages. The alternation -SI / -TI / -JA-SI, if it is morphology, favours D2; if it is local orthography, it decides nothing.

6.3 The closings (E and F)

Both are invariable, come last and occur at six sanctuaries. Neither ever carries a logogram, which rules out their being a second object of offering; their invariance rules out their being names. Two functions remain compatible and cannot be separated with this corpus: a fixed closing formula, or an adverb of time or place. One datum is informative: F can occur without either C or D (VRY Za 1, IO Za 15), so the closing depends neither on the deity nor on the offering, but on the inscription itself.

7. DISCUSSION

What is established. The formula has a template of seven positions whose function is fixed by distribution in five of them; its slots hold the open class with a probability of error of the order of 10^{-4} ; its opening is a root with a paradigm whose variation is local and not grammatical; three morphological elements separate by distribution; and one position, that of the offering, is fixed by internal evidence of the corpus itself.

What is not established, and will not be with this corpus: the value of any of the four anchors. The possible functions of each have been enumerated, those contradicting some datum discarded, and two remain per position in the two central cases. The datum that would separate each pair is specified and does not exist: new inscriptions are needed, and which ones would serve is stated.

What this contributes to the debate on the language. Nothing directly, and that should be said. The comparison of templates is one of structure and not of words, so it favours no linguistic affiliation: a community can adopt the ritual structure of its neighbours while speaking another language, as indeed happens throughout the ancient Mediterranean. What it does provide is a criterion for ordering the existing proposals: a reading proposal that requires a Semitic or Greek template has the structure measured here against it, and one that requires the Anatolian or the Egyptian does not. The criterion decides nothing, but it discards.

And what it contributes to method. In a corpus without a bilingual, the function of each position of a formula is measurable and the number of possible readings can be bounded. What is not measurable is the value of the words. That distinction is what this paper seeks to establish, because a century of proposals has treated the two as one task. ## Acknowledgements

The corpus was consulted in the LinearA Explorer (R. Hogan) and verified sign by sign against SigLA (E. Salgarella and S. Castellan) and GORILA; the sign readings of the inscriptions discussed were checked on the pages of GORILA IV. A language model (Claude, Anthropic) assisted with the analysis and the drafting under the author's direction; the author is responsible for all claims. Code and derived tables are available at github.com/BiomeMakers/kuro.

References

Acedo A. 2026a, Thirteen candidate relatives of Minoan measured against one null, and why none passes, preprint, <https://github.com/BiomeMakers/kuro>.

Acedo A. 2026b, kuro: null models, calibration and confounders for small epigraphic corpora, software, <https://github.com/BiomeMakers/kuro>.

Barta W. 1968, *Aufbau und Bedeutung der altägyptischen Opferformel* (Ägyptologische Forschungen 24), Glückstadt.

Davis B. 2013, Syntax in Linear A: the word-order of the “libation formula”, *Kadmos* 52, 35-52.

Davis B. 2014, *Minoan Stone Vessels with Linear A Inscriptions* (Aegaeum 36), Leuven, Liège.

de Roos J. 2007, *Hittite Votive Texts* (PIHANS 109), Leiden.

Duhoux Y. 1992, Variations, formules et texte dans les inscriptions votives en linéaire A, *Cretan Studies* 3, 65-88.

Facchetti G.M., Negri M. 2003, *Creta minoica. Sulle tracce delle lingue e delle culture del Mediterraneo* (Biblioteca dell'Archivum Romanicum, serie II, 55), Florence.

Finkelberg M. 1990-91, Minoan inscriptions on libation vessels, *Minos* 25-26, 43-85.

Gardiner A.H. 1957, *Egyptian Grammar*, 3rd ed., Oxford.

Godart L., Olivier J.-P. 1976-1985, *Recueil des inscriptions en linéaire A* (Études crétoises 21), I-V, Paris.

Hawkins J.D. 2000, *Corpus of Hieroglyphic Luwian Inscriptions*, I. *Inscriptions of the Iron Age*, Berlin, New York.

Hogan R. 2022, *LinearA Explorer*, <https://lineara.xyz>. Access date: 11 September 2026.

Karnava A. 2016, On sacred vocabulary and religious dedications: the Linear A “libation formula”, in Alram-Stern E., Blakolmer F., Deger-Jalkotzy S., Laffineur R., Weilharter J. (eds), *Metaphysis. Ritual, Myth and Symbolism in the Aegean Bronze Age* (Aegaeum 39), Leuven, Liège, 345-355.

Lazzarini M.L. 1976, Le formule delle dediche votive nella Grecia arcaica, *MemLinc* ser. 8, 19, 47-354.

Pardee D. 2002, *Ritual and Cult at Ugarit* (Writings from the Ancient World 10), Atlanta.

Petrakis V., Steele P.M. 2025, A-SA-SA(-RA-ME), in Salgarella E., Petrakis V. (eds), *The Wor(l)ds of Linear A: Interdisciplinary Approaches to Documents and Inscriptions of a Cretan Bronze Age Script* (AURA Supplement 15), Athens.

Schweitzer S.D. 2021, AES – Ancient Egyptian Sentences. Corpus of Ancient Egyptian sentences for corpus-linguistic research (CC BY-SA 4.0), github.com/simondschweitzer/aes.

Salgarella E., Castellan S. 2020, *SigLA. The Signs of Linear A: a palaeographical database*, <https://sigla.phis.me>. Access date: 11 September 2026.

Salgarella E., Petrakis V. (eds) 2025, *The Wor(l)ds of Linear A: Interdisciplinary Approaches to Documents and Inscriptions of a Cretan Bronze Age Script* (AURA Supplement 15), Athens.

Schweitzer S.D. 2021, AES – Ancient Egyptian Sentences. Corpus of Ancient Egyptian sentences for corpus-linguistic research, <https://github.com/simondschweitzer/aes> (CC BY-SA 4.0). Access date: 12 September 2026.

Sommerschield T., Assael Y., Pavlopoulos J., Stefanak V., Senior A., Dyer C., Bodel J., Prag J., Androutsopoulos I., de Freitas N. 2023, Machine Learning for Ancient Languages: A Survey, *Computational Linguistics* 49, 1-44.

Steele P.M., Meißner T. 2017, From Linear B to Linear A: the problem of the backward projection of sound values, in Steele P.M. (ed.), *Understanding Relations Between Scripts. The Aegean Writing Systems*, Oxford, Philadelphia, 93-110.

Valério M. 2007, “Diktaian Master”: a Minoan predecessor of Diktaian Zeus in Linear A?, *Kadmos* 46, 3-14.

Younger J.G. 2024, *Linear A Texts and Inscriptions in Phonetic Transcription*, <http://people.ku.edu/~jyounger/LinearA/>. Access date: 11 September 2026.

Alberto Acedo Biome Makers Inc., Valladolid, Spain [correo]

Keywords: Linear A, libation formula, votive inscriptions, null models, textual restoration.